# Class 10 Chemistry — Chapter 3: Metals and Non-Metals

**Simple & Detailed NCERT Notes (Exam-Ready)**

## 1. Introduction

- Out of about 118 known elements, roughly **92 occur naturally**. Most of them are **metals**.
- **Metals** are elements that are usually hard, shiny (lustrous), malleable, ductile, sonorous and good conductors of heat and electricity.
  - Examples: Iron (Fe), Copper (Cu), Aluminium (Al), Gold (Au), Silver (Ag), Sodium (Na), etc.
- **Non-metals** are elements that are generally soft (or gases), dull, brittle, non-sonorous and poor conductors of heat and electricity.
  - Examples: Oxygen (O), Nitrogen (N), Carbon (C), Sulphur (S), Chlorine (Cl), Phosphorus (P), etc.
- Metals and non-metals show very different **physical** and **chemical** properties. We will study them one by one.

## 2. Physical Properties of Metals

| Property | Meaning | Example |
|---|---|---|
| **Lustre (shininess)** | Freshly cut metals shine | Gold, silver, copper |
| **Hardness** | Most metals are hard | Iron, copper |
| **Malleability** | Can be beaten into thin sheets | Gold, aluminium foil |
| **Ductility** | Can be drawn into thin wires | Copper wires, gold wires |
| **Sonority** | Produce ringing sound when struck | Metals used in bells |
| **Conductivity** | Good conductors of heat & electricity | Silver & copper are the best |

### ⚠ Exceptions (very important for exams)

- **Mercury (Hg)** is a **liquid** at room temperature.
- **Sodium (Na)** and **Potassium (K)** are so **soft** that they can be cut with a knife.
- **Lead (Pb)** is not very hard; **zinc (Zn)** is brittle at room temperature.
- Mercury is a **poor conductor** of heat compared to other metals (but still a conductor of electricity).

## 3. Physical Properties of Non-Metals

- Generally exist as **solids or gases** at room temperature (**Bromine (Br)** is the only liquid non-metal).
- They are **dull** (no lustre), **brittle** (break into pieces when hammered), **non-sonorous**, and **poor conductors** of heat and electricity (bad conductors = insulators).

### ⚠ Exceptions

- **Diamond** (a form of carbon) is the **hardest natural substance** and has a high melting point.
- **Graphite** (a form of carbon) is a **good conductor of electricity**.
- **Iodine** is a non-metal but has a **shiny (lustrous)** appearance.

## 4. Chemical Properties of Metals

### 4.1 Reaction with Oxygen (Burning in Air)

Metals react with oxygen to form **metal oxides**. Metal oxides are generally **basic** in nature.

```
Metal + Oxygen → Metal oxide

2Na + O₂  → 2Na₂O        (sodium oxide — very reactive, stored in kerosene)
2Mg + O₂  → 2MgO         (magnesium burns with a dazzling white light)
4Al + 3O₂ → 2Al₂O₃       (aluminium forms a protective oxide layer)
3Fe + 2O₂ → Fe₃O₄        (iron filings burn in air; Fe₃O₄ = FeO·Fe₂O₃)
2Cu + O₂  → 2CuO         (black copper(II) oxide forms on heating)
2Zn + O₂  → 2ZnO
```

- **Reactivity with oxygen order:** Na > Mg > Al > Zn > Fe > Cu (metals higher in the reactivity series react more vigorously).
- **Amphoteric oxides:** $Al_2O_3$ and ZnO are special — they react with **both acids and bases**.

```
Al₂O₃ + 6HCl → 2AlCl₃ + 3H₂O      (acts as a base)
Al₂O₃ + 2NaOH + 3H₂O → 2Na[Al(OH)₄] (acts as an acid)
```

### 4.2 Reaction with Water

```
2Na + 2H₂O → 2NaOH + H₂↑ + heat   (very vigorous, may catch fire)
2K  + 2H₂O → 2KOH  + H₂↑ + heat   (even more vigorous than sodium)
Ca  + 2H₂O → Ca(OH)₂ + H₂↑        (less vigorous than Na/K)
Mg  + 2H₂O (hot) → Mg(OH)₂ + H₂↑
Mg  + H₂O (steam) → MgO + H₂↑
2Al + 3H₂O (steam) → Al₂O₃ + 3H₂↑
3Fe + 4H₂O (steam) → Fe₃O₄ + 4H₂↑
Zn  + H₂O (steam) → ZnO + H₂↑
```

- Metals like **Pb, Cu, Ag, Au** do **not** react with water (or steam).
- Metals above hydrogen in the reactivity series can displace hydrogen from water/acids.

### 4.3 Reaction with Dilute Acids

```
Metal + dilute acid → Salt + Hydrogen gas

Mg + 2HCl → MgCl₂ + H₂↑
2Al + 6HCl → 2AlCl₃ + 3H₂↑
Zn + 2HCl → ZnCl₂ + H₂↑
Fe + 2HCl → FeCl₂ + H₂↑     (Fe gives Fe²⁺ salts with dilute acids)
```

- Only metals **above hydrogen** in the reactivity series displace $H_2$ from dilute acids. Cu, Hg, Ag, Au do not.
- **Note:** Nitric acid ($HNO_3$) is a strong oxidising agent — it usually gives oxides of nitrogen instead of $H_2$ (except **very dilute $HNO_3$ with Mg/Mn** gives $H_2$).
- **Aqua regia** = 3 parts conc. HCl + 1 part conc. $HNO_3$. It can **dissolve gold and platinum** (the most noble metals).

### 4.4 Reaction with Salt Solutions (Displacement Reactions)

A **more reactive metal displaces a less reactive metal** from its salt solution.

```
Fe + CuSO₄ → FeSO₄ + Cu        (blue colour of CuSO₄ fades; copper deposits on iron)
Zn + CuSO₄ → ZnSO₄ + Cu
Cu + 2AgNO₃ → Cu(NO₃)₂ + 2Ag   (silver shines on copper)
```

### ★ The Reactivity Series (memorise this!)

```
K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Pt > Au
Most reactive ──────────────────────────────────────────► Least reactive
```

- A metal **higher** in the series displaces a metal **lower** in the series from its salt solution.
- Metals above **hydrogen [H]** can displace hydrogen from dilute acids and water.

### 4.5 Reaction with Chlorine

Metals react with chlorine to form **ionic (electrovalent) chlorides**.

```
2Na + Cl₂ → 2NaCl
Mg + Cl₂  → MgCl₂
2Al + 3Cl₂ → 2AlCl₃
Zn + Cl₂  → ZnCl₂
2Fe + 3Cl₂ → 2FeCl₃   (iron(III) chloride)
Cu + Cl₂  → CuCl₂
```

### 4.6 Reaction with Hydrogen

Only the **most reactive metals** (like Na, Ca) combine with hydrogen to form **metal hydrides** (ionic compounds).

```
2Na + H₂ → 2NaH
Ca + H₂  → CaH₂
```

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## 5. Chemical Properties of Non-Metals

- Non-metals react with oxygen to form **non-metal oxides**, which are generally **acidic** in nature.

```
C + O₂    → CO₂
S + O₂    → SO₂
4P + 5O₂  → 2P₂O₅
```

- Non-metals do **not** displace hydrogen from dilute acids (they are below H in the reactivity series).
- Reaction with chlorine: non-metals form **covalent chlorides**.

```
H₂ + Cl₂ → 2HCl   (covalent, not ionic)
```

- Reaction with hydrogen:

```
H₂ + S  → H₂S
N₂ + 3H₂ ⇌ 2NH₃   (Haber process, with iron catalyst)
```

- Special: Chlorine reacts with water:

```
Cl₂ + H₂O → HCl + HOCl   (bleaching action)
```

### Metals vs Non-Metals — Chemical Properties (quick table)

| Property | Metals | Non-metals |
|---|---|---|
| Reaction with $O_2$ | Form **basic** oxides | Form **acidic** oxides |
| Reaction with acids | Displace $H_2$ (if above H) | Do not displace $H_2$ |
| Reaction with salt solutions | Displace less reactive metals | No displacement |
| Reaction with $Cl_2$ | Ionic chlorides | Covalent chlorides |
| Reaction with $H_2$ | Only reactive metals form hydrides | Many non-metals form hydrides ($H_2S$, HCl, $NH_3$) |

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## 6. Ionic (Electrovalent) Bonding

- When a **metal** (low electronegativity) reacts with a **non-metal** (high electronegativity), the metal **loses** electron(s) and the non-metal **gains** them.
- The oppositely charged ions attract each other by strong **electrostatic force → ionic bond**.
- Example — formation of NaCl:

```
Na (2,8,1) → Na⁺ + e⁻        (loses 1 electron)
Cl (2,8,7) + e⁻ → Cl⁻        (gains 1 electron)
Na⁺ + Cl⁻ → NaCl             (ionic/electrovalent bond)
```

### Properties of Ionic Compounds

- **Hard and brittle** (strong electrostatic attraction; break on hammering).
- **High melting and boiling points** (strong bonds need lots of energy).
- **Soluble in water**, insoluble in organic solvents like kerosene/petrol.
- **Conduct electricity in molten state or in aqueous solution** (ions are free to move), but **not in solid state** (ions are fixed in the lattice).

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## 7. Occurrence of Metals

- The **earth's crust** and **sea water** are the sources of metals.
- **Minerals:** naturally occurring substances from which metals can be extracted (e.g., bauxite, haematite).
- **Ore:** a mineral from which a metal can be extracted **profitably and conveniently** (all ores are minerals, but all minerals are not ores).
- **Gangue:** the unwanted earthy impurities (sand, clay, etc.) mixed with the ore.
- **Metallurgy:** the entire process of extracting metals from their ores and refining them.

### Enrichment of Ore (removing gangue) — based on properties:

| Method | Principle | Used for |
|---|---|---|
| **Gravity separation / Levigation** | Ore is heavier than gangue; washed with water | Oxide ores (haematite) |
| **Froth flotation** | Sulphide ores are wetted by oil, gangue by water | $Cu_2S$, PbS, ZnS ores |

| Method | Principle | Used for |
|---|---|---|
| **Magnetic separation** | Magnetic ore is separated by a magnet | Iron ores (magnetite) |
| **Leaching** | Ore dissolves in a suitable solvent (e.g., NaOH for bauxite → $NaAlO_2$) | Bauxite ($Al_2O_3 \cdot 2H_2O$) |

## 8. Extraction of Metals (based on the Reactivity Series)

The extraction method depends on where the metal sits in the reactivity series.

### (a) Metals of LOW reactivity (bottom of series: Hg, Au, Ag, Pt)

- Extracted by **heating alone** (roasting/decomposition).

```
2HgS + 3O₂ → 2HgO + 2SO₂     (roasting of cinnabar)
2HgO → 2Hg + O₂↑              (heating mercuric oxide)
```

### (b) Metals of MEDIUM reactivity (middle: Zn, Fe, Pb, Cu)

- Ore is first converted to oxide, then **reduced with carbon (coke)** — called **smelting**.

```
ZnCO₃ → ZnO + CO₂            (calcination – heating in limited air)
ZnS + 3O₂ → 2ZnO + 2SO₂      (roasting – heating in excess air)
ZnO + C → Zn + CO↑            (reduction with coke)
Fe₂O₃ + 3C → 2Fe + 3CO↑
PbO + C → Pb + CO↑
```

- **Thermite (aluminothermic) process:** for metals like Fe, Mn, Cr — reduced by aluminium powder; huge heat released (~2000 °C).

```
Fe₂O₃ + 2Al → Al₂O₃ + 2Fe + heat   (used to weld railway tracks)
```

### (c) Metals of HIGH reactivity (top: K, Na, Ca, Mg, Al)

- Cannot be reduced by carbon — extracted by **electrolysis of their molten compounds**.

```
2NaCl (molten) → 2Na (cathode) + Cl₂↑ (anode)     [Downs cell]
2Al₂O₃ (molten, in cryolite Na₃AlF₆) → 4Al (cathode) + 3O₂↑ (anode)  [Hall–Heroult process]
```

- Electrolysis of molten NaCl: $Na^+$ goes to cathode, $Cl^-$ to anode.
- For aluminium: $Al_2O_3$ is dissolved in **molten cryolite** to lower the melting point (~1000 °C) and increase conductivity.

### Refining of Metals (Electrolytic Refining)

- Impure metal = **anode**, pure metal = **cathode**, acidified salt solution of the metal = **electrolyte**.
- On electrolysis, pure metal deposits on the cathode; impurities settle below the anode as **anode mud**.
- Example — refining of copper:

```
At cathode: Cu²⁺ + 2e⁻ → Cu (pure)
At anode:   Cu → Cu²⁺ + 2e⁻ (impure Cu dissolves)
```

## 9. Corrosion

- **Corrosion:** the slow, gradual destruction of metals by reaction with air, moisture, acids, gases in the environment.
- **Rusting of iron** is the most common example:

```
4Fe + 3O₂ + xH₂O → 2Fe₂O₃·xH₂O   (rust = hydrated iron(III) oxide, reddish-brown)
```

  - Conditions needed: **both oxygen AND moisture (water)**.
  - Rust is soft and porous, so it does not protect the metal underneath.
- Other examples:
  - **Copper** → green coating of basic copper carbonate: $CuCO_3 \cdot Cu(OH)_2$ (patina)
  - **Silver** → black coating of silver sulphide: $Ag_2S$ (tarnishing)

### Prevention of Corrosion

- **Painting / oiling / greasing** — keeps air and moisture away.
- **Galvanisation** — coating iron with a layer of **zinc (Zn)** (e.g., galvanised pipes). Zinc protects even when scratched.
- **Chrome plating / tin plating** — coating with chromium or tin.
- **Alloying** — making stainless steel (Fe + Cr + Ni) which resists rusting.
- **Anodising** — for aluminium (thick oxide layer by electrolysis).

Note: ⚠ Corrosion of iron is a serious problem — billions of rupees are lost every year due to rusting of bridges, ships, cars, etc.

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## 10. Alloys

- An **alloy** is a **homogeneous mixture of two or more metals**, or a metal with a non-metal.
- Alloys **cannot be separated into their components by physical methods**.
- Made to **improve properties** (hardness, strength, resistance to corrosion, lower melting point, better appearance).

### Common Alloys (memorise composition!)

| Alloy | Composition | Special property / use |
|---|---|---|
| **Brass** | Cu + Zn (70:30) | Sonorous — musical instruments |
| **Bronze** | Cu + Sn (88:12) | Hard, corrosion-resistant — statues, coins |
| **Solder** | Pb + Sn (50:50) | Low melting point — soldering electrical wires |
| **Pewter** | Sn + Cu + Sb + Bi | Low melting point — utensils |
| **Duralumin** | Al + Cu + Mg + Mn | Light and strong — aircraft parts |
| **Stainless steel** | Fe + Cr + Ni (+ C) | Does not rust — kitchen utensils, cutlery |
| **German silver** | Cu + Zn + Ni | Looks like silver — decorative items (no actual silver!) |
| **Amalgam** | Hg + another metal (e.g., Na, Ag, Sn) | Used in dentistry (silver amalgam for fillings) |

| Alloy | Composition | Special property / use |
|---|---|---|
| **24-carat gold** | Pure gold (99.9%) | Too soft for jewellery |
| **22-carat gold** | Au + Cu/Ag | Harder — used for jewellery |

- **Why alloys?** Pure metals are often too soft (gold), rust easily (iron), or are reactive — alloying makes them harder, stronger, more durable and rust-resistant.

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## 11. Quick Revision — One-Page Summary

### ★ Most Important Points

- Metals are lustrous, malleable, ductile, sonorous, good conductors. Exceptions: Hg (liquid), Na/K (soft), Pb (soft).
- Non-metals are dull, brittle, poor conductors. Exceptions: diamond (hardest), graphite (conducts), iodine (lustrous), Br (liquid).
- Metal oxides are **basic**; non-metal oxides are **acidic**; $Al_2O_3$ and ZnO are **amphoteric**.
- **Reactivity series:** K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Pt > Au.
- Metals above H displace $H_2$ from dilute acids; more reactive metal displaces less reactive metal from its salt solution.
- Ionic bond = transfer of electrons; ionic compounds conduct electricity only when molten or dissolved in water.
- Extraction: low reactivity → heating; medium → reduction with carbon; high → electrolysis of molten compounds; then electrolytic refining.
- Rusting needs **$O_2$ + $H_2O$**; prevented by painting, galvanisation (Zn), alloying, etc.
- Alloys are homogeneous mixtures of metals (or metal + non-metal) made to improve properties.

### ★ Important Equations to Remember

$2Na + O_2 \rightarrow 2Na_2O$
$2Mg + O_2 \rightarrow 2MgO$
$4Al + 3O_2 \rightarrow 2Al_2O_3$
$3Fe + 2O_2 \rightarrow Fe_3O_4$
$2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$
$3Fe + 4H_2O\ (steam) \rightarrow Fe_3O_4 + 4H_2\uparrow$
$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$
$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
$2Na + Cl_2 \rightarrow 2NaCl$
$Na^+ + Cl^- \rightarrow NaCl$
$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$
$ZnO + C \rightarrow Zn + CO\uparrow$
$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe + heat$
$2NaCl\ (molten) \rightarrow 2Na + Cl_2\uparrow$
$4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O\ (rust)$

### ★ Quick Differences

| Metals | Non-metals |
|---|---|
| Hard, lustrous, malleable, ductile | Soft/dull, brittle (if solid) |
| Good conductors of heat & electricity | Poor conductors (insulators) |
| Form basic oxides | Form acidic oxides |
| Form ionic compounds with non-metals | Form covalent compounds with non-metals |
| Lose electrons (electropositive) | Gain electrons (electronegative) |

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## 12. Exam Tips

- **Very short answer (1 mark):** Define alloy / gangue / corrosion / roasting / calcination / galvanisation; give exceptions of physical properties.
- **Short answer (2–3 marks):** Why do ionic compounds conduct electricity in molten state but not in solid? Why is sodium stored in kerosene? Write the thermite reaction and its use. Difference between metals and non-metals.
- **Long answer (5 marks):** Explain extraction of metals from ores with a flow chart (ore → enrichment → oxide → metal → refining). Explain rusting with an activity/experiment and methods of prevention.
- **Activity to remember:** Put iron nails in (a) dry air, (b) boiled water + oil, (c) water — only (c) rusts → proves both air and water are needed.

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Notes prepared as per NCERT Class 10 Science (Chemistry) Chapter 3 — "Metals and Non-Metals". Best of luck for your exam!